TECHNICAL DESIGN DOCUMENT (TDD)

Jewellery Gold Accountability, Reconciliation & Visual Slip OCR Platform

Target Enterprise	Company A — Coimbatore Jewellery Chain (5 Showrooms, 1 Central Workshop, ~40 Karigars, ■220–260 Cr Revenue)
GitHub Repository	**https://github.com/Ayush5424/jewellery-gold-accountability**
Tech Stack	Spring Boot 3.3 (Java 21) + React 18 (TypeScript) + FastAPI (Python 3.13) + Tesseract OCR + PostgreSQL 16+
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1. Project Title & GitHub Repository URL

Project: Jewellery Gold Accountability, Reconciliation & Anomaly Detection Platform

Public Repository: https://github.com/Ayush5424/jewellery-gold-accountability

The platform is an enterprise-grade accountability solution built specifically to solve physical gold tracking, wastage discrepancy, and multi-location inventory reconciliation challenges in Coimbatore jewellery manufacturing.

2. Executive Summary & Problem Statement

Company A operates five retail showrooms and one centralized workshop in Coimbatore, Tamil Nadu, employing ~40 artisan karigars and 180 total staff with an annual revenue of ■220–260 crore. Previously, raw bullion issuance, scrap returns, and finished jewellery intake were tracked through manual paper notebooks and disconnected Excel sheets. Monthly reconciliation required six senior personnel working for three full days, during which shrinkage or karigar benchmark drift remained undetected for weeks.

This platform establishes end-to-end mathematical accountability: every gram of issued gold is tracked through an immutable append-only transaction ledger, digital weighing scale and karat-meter slips are extracted via genuine pixel-level OCR with mandatory human-in-the-loop confirmation, and stock journals are automatically exported to Tally ERP 9 / Tally Prime.

3. Client Requirements & Scope

The system directly addresses all explicit requirements of the Isocode Labs assignment:

- **Showrooms & Workshop Coverage:** Real-time synchronization across 5 retail showrooms (Cross Cut Road, Gandhipuram, RS Puram, Town Hall, Peelamedu) and 1 centralized workshop.
- **Karigar Gold Balance Tracking:** Complete ledger history of issued gold, finished jewellery returned, cutting scrap recovered, and bench filings collected.
- **Loss & Wastage Reconciliation:** Enforce category-specific wastage allowances and flag abnormal loss drift across consecutive job orders.
- **AI Visual Slip Reader:** Extract weight and purity from digital scale photos and karat-meter printouts, blocking save operations if manual entry disagrees with optical readings.
- **Tally ERP Integration:** Automated export of daily stock movements into Tally XML format.

4. Solution Overview

The solution is architected as a cohesive, three-tier application with clear separation of concerns. Fine gold calculations are computed strictly in pure gold mass (Gross Weight × Purity Touch / 100) using Java's `BigDecimal` to completely eliminate floating-point drift. A standalone Python FastAPI microservice executes multi-pass Tesseract OCR and OpenCV preprocessing without any regex heuristics on raw binary bytes. Role-tailored frontend dashboards provide intuitive interfaces for Finance Heads, Issue Clerks, and Karigars with Tamil localization.

5. System Architecture

The platform consists of four decoupled runtime components communicating over secure HTTP REST protocols. It supports both multi-container Docker Compose orchestration and native multi-process local execution:

Component	Technology	Port	Operational Role & Execution Model
Frontend SPA	React 18, TypeScript, Vite, Tailwind CSS	5173	Role-based dashboards, offline draft caching, image preview, Tamil i18n. Runs locally via Vite dev server or compiled Nginx bundle.

Backend API	Spring Boot 3.3, Java 21, Spring Data JPA	8080	Authoritative accounting arithmetic, immutable ledger, Tally XML export. Runs via Maven spring-boot:run or containerized JVM.
AI OCR Service	FastAPI, Python 3.13, Tesseract 5, OpenCV	8000	Image decoding, Laplacian blur check, CLAHE, Otsu LCD crop, OCR extraction. Runs locally via Uvicorn asynchronous server.
Database	PostgreSQL 16+ Relational DB	5432	ACID transactions, foreign keys, unique constraint indexes, audit trail. Runs via local Windows service or PostgreSQL container.

6. Technology Stack & Rationale

Layer	Technology	Version	Selection Rationale
Language (Core)	Java (OpenJDK)	21 LTS	Strong static typing, enterprise maintainability, and precise BigDecimal arithmetic.
Framework (API)	Spring Boot	3.3.5	Production-grade REST ecosystem, Spring Security, JPA Hibernate, HikariCP.
AI & Vision	Python & FastAPI	3.13	Async I/O for image handling, native OpenCV computer vision, Tesseract bindings.
OCR Engine	Tesseract OCR	5.4.1	Genuine pixel-level neural optical character recognition; local execution with zero cloud cost.
Frontend Framework	React + Vite	18.3 / 5.4	Rapid HMR, component isolation, responsive rendering, zero bundle bloat.
Styling & Icons	Tailwind + Lucide	3.4 / 0.344	Cohesive enterprise design system with professional dark/light palette.
Relational DB	PostgreSQL	16+	Robust ACID guarantees, transactional isolation, and index-backed referential integrity.

7. Database Schema & Entity Relationships

The relational model is designed for strict auditability, append-only immutability, and referential integrity:

Entity / Table	Primary Key	Key Attributes & Foreign Keys	Domain Invariants & Constraints
`app_user`	id (BIGINT)	username (UNIQUE), password_hash, full_name, role, karigar_id (FK)	BCrypt hashed passwords; roles: ROLE_FINANCE_HEAD, ROLE_ISSUE_CLERK, ROLE_KARIGAR.
`karigars`	id (BIGINT)	code (KRG-xxx, UNIQUE), name, phone, bench_location, completed_jobs_count	Artisans across Coimbatore showrooms/workshop. Tracks rolling loss averages.
`job_orders`	id (BIGINT)	job_number (JOB-yyyy-xxx, UNIQUE), karigar_id (FK), category, agreed_wastage_pct, status	Status lifecycle: ACTIVE -> READY_FOR_RECONCILIATION -> COMPLETED. Wastage % snapshotted.
`gold_ledger_entry`	id (BIGINT)	job_order_id (FK), karigar_id (FK), transaction_type, gross_weight, purity, fine_weight, voucher_number	Immutable append-only ledger with unique voucher sequencing and idempotency keys.
`karigar_wastage_allowance`	id (BIGINT)	karigar_id (FK), item_category, default_wastage_pct, updated_by, updated_at	Unique constraint on (karigar_id, item_category). Persistent matrix configuration.
`loss_alert`	id (BIGINT)	job_order_id (FK), karigar_id (FK), alert_type, actual_loss, allowed_loss, excess_loss, status	Alert types: EXCESS_OVER_ALLOWANCE, HISTORICAL_DRIFT. Status: PENDING_REVIEW, REVIEWED_APPROVED, ESCALATED.

8. Gold Accounting Calculations with Worked Examples

All mathematical computations in `GoldAccountingCalculator.java` use Java `BigDecimal` with three decimal places (milligrams, scale 3) and `RoundingMode.HALF_UP`. Pure fine gold mass is computed independently of physical alloy bulk weight:

- **Fine Gold Weight Formula:** `fineWeight = grossWeight × (purity / 100)` (scale 3, HALF_UP)
- **Recovered Fine Gold Formula:** `recoveredFineWeight = jewelleryFine + scrapFine + filingsFine`
- **Actual Loss Formula:** `actualLoss = issuedFineWeight - recoveredFineWeight`
- **Allowed Loss Formula:** `allowedLoss = issuedFineWeight × (allowedWastagePercent / 100)` (scale 3, HALF_UP)
- **Excess Loss Formula:** `excessLoss = actualLoss - allowedLoss`
- **Loss Percentage Formula:** `lossPct = (actualLoss × 100) / issuedFineWeight` (scale 2, HALF_UP)

Comprehensive Benchmark Worked Example (Inputs: 120.000g Issued @ 91.60% Purity, 1.20% Allowed Wastage):

Accounting Line Item	Physical Gross Weight (g)	Purity Touch (%)	Fine Gold Mass (g)	Accounting Computation & Rules
Raw Bullion Issued	120.000 g	91.60 %	**109.920 g**	120.000 × (91.60 / 100) = 109.920 g Fine
Receipt: Finished Jewellery	112.450 g	91.60 %	**103.004 g**	112.450 × 91.60% = 103.0042 -> 103.004 g
Receipt: Cutting Scrap	3.300 g	91.60 %	**3.023 g**	3.300 × 91.60% = 3.0228 -> 3.023 g
Receipt: Bench Filings	0.450 g	91.60 %	**0.412 g**	0.450 × 91.60% = 0.4122 -> 0.412 g
Total Accounted Return	**116.200 g**	91.60 %	**106.439 g**	103.004 + 3.023 + 0.412 = 106.439 g Fine
Actual Loss	**3.800 g** (Gross diff)	N/A	**3.481 g** (Actual fine loss)	109.920 - 106.439 = 3.481 g Fine (3.17%)
Allowed Wastage (1.20%)	**1.440 g** (Gross equiv)	N/A	**1.319 g** (Allowed fine loss)	109.920 × 1.20% = 1.31904 -> 1.319 g Fine
Excess Loss (Unexplained)	**2.360 g** (Gross excess)	N/A	**2.162 g** (Excess fine loss)	3.481 - 1.319 = 2.162 g Fine (1.97% excess)
Anomaly Alert Evaluation	Excess loss: 2.162 g	N/A	1.97% > Tolerance (0.10%)	**EXCESS_OVER_ALLOWANCE Alert Flagged**

9. Gold Issue and Receipt Lifecycle

Gold issuance and return transitions through strict state validation in `JobOrderLifecycleService`:

1. **Job Creation & Raw Bullion Issue:** Issue clerk registers artisan, selects item category, inputs raw gold weight and purity touch, generating an active Job Order (`JOB-2026-xxx`) and an initial `ISSUE_BULLION` transaction voucher.
2. **Interim Deliveries:** Artisans return finished items in single or multiple batches (`RECEIPT_JEWELLERY` with HUID). Each receipt reduces outstanding artisan custody in real time.
3. **Scrap & Sweep Collection:** Cutting scraps (`RECEIPT_SCRAP`) and collected bench dust (`RECEIPT_FILINGS`) are weighed on precision scales and logged.
4. **Reconciliation Transition:** Once manufacturing completes, clerk flags the job as `READY_FOR_RECONCILIATION`. Further gold issues on this job are blocked.
5. **Finance Head Closure:** Finance Head audits variance. Upon approval, job state moves to `COMPLETED`, closing the artisan balance and updating historical moving averages.

10. Wastage Allowance Configuration

To reflect artisan skill variance and product intricacy across Coimbatore workshops, the platform provides a persistent category wastage matrix (`KarigarWastageAllowance`):

- **Item Categories Supported:** CHAIN (1.20% default), BANGLE (1.00%), RING (1.00%), NECKLACE (1.50%), TEMPLE_JEWELLERY (1.80%), EARRINGS (1.20%), COIN (0.20%).
- **Matrix Persistence:** Stored in PostgreSQL with unique constraint on `(karigar_id, item_category)`. Managed exclusively by Finance Heads.
- **Snapshot Immutability:** When a Job Order is created, the configured wastage percentage is permanently copied into `JobOrder.agreed_wastage_pct`. Future matrix modifications never alter historical reconciliations.
- **Authorized Overrides:** The Issue Clerk can adjust the agreed wastage on specific custom designs, requiring an audit justification that is logged in transaction remarks.

11. Ledger & Stock Reconciliation

Gold accounting is governed by an immutable append-only transaction ledger with unique voucher sequencing and idempotency keys:

- **Critical Domain Rule — Unclosed Work is WIP:** In jewellery manufacturing, open job orders represent active Work-in-Progress (WIP). Unfinished gold in karigar custody is strictly recorded as active workshop asset, **never automatically treated as loss or shrinkage** until reconciliation finalization.
- **Ledger Immutability & Double-Entry Mapping:** Transactions (`ISSUE_BULLION`, `RECEIPT_JEWELLERY`, `RECEIPT_SCRAP`, `RECEIPT_FILINGS`) are permanently logged. Each transaction maps to balanced double-entry debits and credits in the exported Tally XML Stock Journal.

- **Showroom & Workshop Balances:** The Daily Stock Statement aggregates vault stock, workshop WIP, and showroom finished inventory across all five retail branches.
- **Tally XML Stock Journal:** Generates standard Tally Prime / ERP 9 XML stock journal vouchers to debit raw gold inventory and credit finished jewellery stock upon reconciliation.

12. Authentication & Role-Based Access Control

The platform enforces stateless JWT authentication with strict role-based access control (RBAC):

Role Name	Assigned Personas	Access Scope & Security Boundaries
`ROLE_FINANCE_HEAD`	Finance Head (Saravanan R)	Full enterprise visibility: total stock statement, company-wide loss dashboards, anomaly alerts, wastage matrix editor, reconciliation approval, Tally XML export.
`ROLE_ISSUE_CLERK`	Issue Clerk (Ramalingam K)	Operational desk: registers karigars, issues raw gold, logs receipts, uploads scale slips via AI OCR, initiates reconciliation. Blocked from editing company settings.
`ROLE_KARIGAR`	Artisans (Murugesan, Sivakumar)	Strict row-level isolation: Karigar users are bound to their Karigar ID and can **only access their own jobs and personal ledger**. Zero visibility into peers.

13. AI OCR Architecture, Tesseract, Preprocessing & Confidence Handling

The optical reading microservice (`ai-service`) executes genuine pixel-level OCR using Tesseract and OpenCV preprocessing. No heuristics or raw-byte regex decoding are used:

1. **Image Validation & Decoding:** Uploaded image bytes are decoded into OpenCV BGR matrices. Invalid formats are rejected.
2. **Laplacian Blur Analysis:** Evaluates focus via `cv2.Laplacian(gray, cv2.CV_64F).var()`. Images with variance below threshold (< 20.0) are flagged as severely blurred, immediately triggering a manual entry warning.
3. **Multi-Pass Preprocessing:** Applies Contrast-Limited Adaptive Histogram Equalization (CLAHE), Otsu binarization for thermal slips, and bitwise inversion for bright LED digits on dark scale screens.
4. **LCD Display Window Crop:** Detects dark digital display bezels, crops the digit region, and upscales by 2.0× using bicubic interpolation before running line-level OCR.
5. **Rotation Correction:** If OCR character count is low, multi-angle evaluation checks 90°, 180°, and 270° orientations.
6. **Confidence Scoring:** Computes confidence based on recognized tokens and blur metrics. Outputs are categorized as `HIGH_CONFIDENCE_DRAFT` (≥ 0.70) or `LOW_CONFIDENCE_MANUAL_REQUIRED` (< 0.70).

14. Human Confirmation & Mismatch-Blocking Workflow

In accordance with Isocode Labs client requirements, AI readings are strictly non-authoritative drafts:

- **Side-by-Side Verification:** The UI displays the original scale photo alongside extracted weight and purity fields.
- **Mandatory Confirmation:** API responses return `requires_human_confirmation: true`. Transactions cannot save without clerk review.
- **Mismatch Hard Blocking Gate:** If the clerk's manually entered weight differs by > 0.001 g or purity differs by > 0.01% from OCR readings, saving is **hard-blocked** (`isBlockedByMismatch = true`).
- **Override Audit Trail:** Unblocking requires entering an explicit override reason, which is permanently appended to the immutable transaction remarks: `[Manual Override: <reason>]`.

15. Frontend Architecture & Important Screens

The React 18 SPA is structured into modular, role-tailored pages built with Vite and Tailwind CSS:

Page / Component	Route	Target User	Functional Capabilities
Finance Dashboard	`/dashboard`	Finance Head	Enterprise inventory summary, active WIP, loss alerts, Tally export.
Issue Clerk Desk	`/clerk`	Issue Clerk	Artisan quick-select, active jobs overview, gold issuance and receipt actions.
Karigar Dashboard	`/karigar`	Artisans	Personal jobs, historical transaction ledger, outstanding gold balance. Tamil toggle.
Issue Gold	`/issue`	Issue Clerk	Bullion issuance form, auto-populated agreed wastage %, fine gold preview.
Receive Gold	`/receive`	Issue Clerk	Receipts for jewellery, scrap, and filings. HUID entry and fine gold computation.
AI Slip Scanner	`/ai-slip`	Issue Clerk	Scale slip photo upload, side-by-side preview, mismatch gate, and confirmation.
Stock Statement	`/statement`	Finance Head	Daily showroom/workshop stock statement with one-click Tally XML export.

Karigar Master	`/karigars`	Issue Clerk / Finance	Artisan registration, contact management, and bench unit assignment.

16. Tamil Localization

To support artisan karigars in Coimbatore workshops, the Master Karigar portal features a real-time English / ■■■■■ language toggle (`KarigarDashboardPage.tsx`):

- **Translated Elements:** Navigation tabs, dashboard cards, action buttons, transaction statuses, and balance labels.
- **Accounting Immutability:** Numeric values, 3-decimal weights (e.g. `12.450 g`), purity percentages (`91.60%`), Job Order numbers, voucher IDs, and HUIDs remain strictly untranslated to prevent audit confusion.
- **State Persistence:** Language preference is stored in browser `localStorage` and retained across sessions.

17. Security & Configuration Management

The system enforces defense-in-depth security across all layers:

- **BCrypt Password Hashing:** Passwords are encrypted using Spring Security's `BCryptPasswordEncoder` with strength factor 10. Cleartext passwords are never stored.
- **Idempotent Seeding:** `DataInitializer.java` loads demo passwords via environment variables (`DEMO_*_PASSWORD`). If a user exists in PostgreSQL, existing password hashes are **never overwritten or reset** on application reboot.
- **Credential Isolation:** Local `.env` is excluded from Git via `.gitignore`. Public repository contains only sanitized `.env.example` with placeholder tokens.
- **Credential Rotation:** Historical evaluation credentials from early development commits were formally rotated in local PostgreSQL; old passwords throw `BadCredentialsException`.

18. Testing Results & Verification Evidence

All modules were rigorously validated across three comprehensive automated test suites and live image fixture evaluations:

- **Backend Spring Boot Suite: 41 / 41 Tests Passing** (0 failures, 0 errors, ~13s runtime). Validates gold accounting formulas, concurrency locks, job lifecycle transitions, idempotent user seeding, and BCrypt security.
- **AI Microservice Pytest Suite: 11 / 11 Tests Passing** (0 failures, ~4s runtime). Validates API endpoints, Karat-grade extraction, blur rejection, and Pydantic schemas.
- **Frontend Production Build: 1,516 Vite modules transformed** with 0 errors in 2.18s.

Photographic Fixture Verification (12 Real Image Samples):

All 12 test fixtures in `test-data/ai-slip-samples/` were submitted to the live FastAPI endpoint (`POST /extract-slip`):

Image Fixture	Category	Expected Weight	Extracted Weight	Expected Purity	Extracted Purity	Result & Behavior
scale_01_clear_12_450g.png	WEIGHING_SCALE	12.450 g	12.450 g	N/A	N/A	**PASS (Exact Numeric Match)**
scale_02_clear_25_680g.png	WEIGHING_SCALE	25.680 g	25.680 g	N/A	N/A	**PASS (Exact Numeric Match)**
scale_03_clear_8_125g.png	WEIGHING_SCALE	8.125 g	8.125 g	N/A	N/A	**PASS (Exact Numeric Match)**
scale_04_angled_15_300g.png	WEIGHING_SCALE	15.300 g	15.300 g	N/A	N/A	**PASS (Angled View Corrected)**
scale_05_glare_45_200g.png	WEIGHING_SCALE	45.200 g	None	N/A	N/A	**PASS (Safe Glare Downgrade)**
scale_06_blurry_obscured.png	WEIGHING_SCALE	30.100 g	None	N/A	N/A	**PASS (Safe Blur Rejection)**
purity_01_clear_22k_9160.png	KARAT_METER	N/A	N/A	91.60 %	91.60 %	**PASS (Exact Touch Match)**
purity_02_clear_18k_7500.png	KARAT_METER	N/A	N/A	75.00 %	75.00 %	**PASS (Exact Touch Match)**
purity_03_clear_20k_8333.png	KARAT_METER	N/A	N/A	83.33 %	83.33 %	**PASS (Exact Touch Match)**

slip_04_dual_weight_purity.png	DUAL_PRINTOUT	50.000 g	50.000 g	91.60 %	91.60 %	**PASS (Dual Exact Match)**
slip_05_rotated_poor_light.png	ROTATED_SLIP	N/A	N/A	91.60 %	91.60 %	**PASS (90° Rotation Handled)**
slip_06_obscured_purity.png	OBSCURED_SLIP	34.250 g	None	N/A	N/A	**PASS (Safe Blur Downgrade)**

19. Assumptions & Design Trade-offs

1. **Deterministic Moving-Average vs Neural Anomaly Model:** The assignment prompt requires detecting karigar loss drift. Rather than introducing an opaque deep learning neural network with high training overhead, a deterministic rolling statistical moving-average was chosen. This ensures 100% auditability and explainability to workshop finance officers.

2. **Local Execution Priority:** In accordance with assignment instructions, live cloud deployment was not performed. The system is designed and documented for local Docker / native multi-process execution.

3. **ACID Relational Model vs Event Sourcing:** PostgreSQL with transactional row locking was selected over complex distributed event sourcing to minimize infrastructure complexity while ensuring strict financial consistency.

4. **Offline Draft Persistence:** Frontend stores unsubmitted slips in `localStorage` to prevent data loss during temporary network disconnects between workshop benches.

20. Limitations & Future Improvements

- **Hardware Scale Serial Connection:** Current system processes scale photos via AI OCR. Future iterations can integrate direct RS-232 / USB scale telemetry for instant automated readings.
- **Mobile Native Camera Application:** Packaging the React SPA as a Progressive Web App (PWA) with direct camera viewfinder capture.
- **Biometric Artisan Sign-Off:** Incorporating optical fingerprint scanners at the issue desk for non-repudiation during raw bullion handover.
- **Multi-Currency Bullion Hedging:** Live integration with MCX gold spot prices to compute real-time rupee financial exposure alongside physical gram weight.